# Practice workbook · Use image size, actual size and magnification

A micrograph or diagram with a measurable length, a stated real size or magnification, or eyepiece and objective lenses.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-MAGNIFICATION

# 1. Question 3 · part b.i · 2 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b.i only.

**3** (a) Describe what is meant by a specialised cell.

**(2)**

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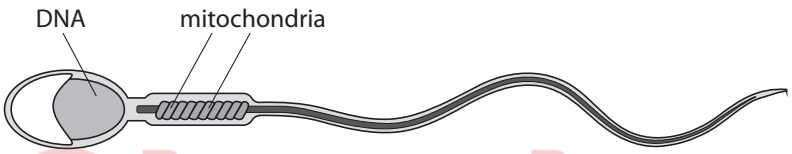
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE




(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

**(2)**

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

**(1)**

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## 1. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

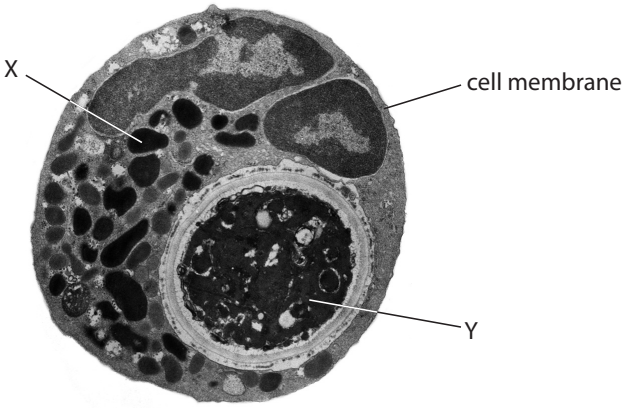
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# 2. Question 2 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

**2** (a) The image is an electron micrograph of a human body cell.




(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

**(2)**

| Part | Name of part | Function |
|---|---|---|
| X | | release energy from glucose |
| Y | nucleus | |

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

**(1)**

- ☒ **A** more cell structures can be seen with greater resolution
- ☒ **B** more cell structures can be seen with less resolution
- ☒ **C** fewer cell structures can be seen with greater resolution
- ☒ **D** fewer cell structures can be seen with less resolution

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## 2. Question 2 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

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(iii) The actual diameter of this human body cell before magnification is 0.05 mm.

Use information from the image to calculate the magnification of the cell.

**(3)**

magnification = .....

(b) Body cells produce carbon dioxide.

Describe how carbon dioxide passes from a body cell into the blood.

**(2)**

.....

.....

.....

.....

**(Total for Question 2 = 8 marks)**

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## 2. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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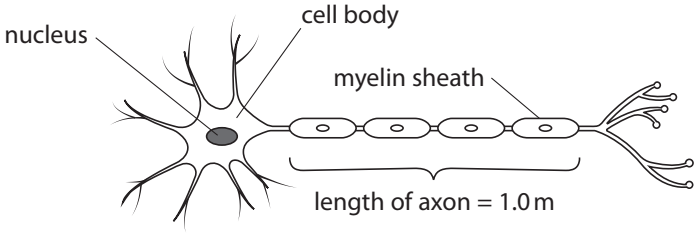
## 3. Question 5 · part a.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

**5** The diagram shows a motor neurone.

not to scale




(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

**(2)**

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diameter = ..... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

**(1)**

.....

.....

(b) Nerve impulses travel at an average speed of $1.2 \times 10^8$ µm per second.

Hormones travel at an average speed of $1.5 \times 10^6$ µm per second.

(i) Explain this difference in speed.

**(2)**

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## 3. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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# 4. Question 3 · part b.i · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.i only.

**3** (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

**(2)**

| | **Steps** |
|---|---|
| | select a suitable objective lens |
| 5 | adjust focusing wheel to obtain a clear image |
| 1 | remove cells from the inside of the cheek |
| | place microscope slide onto the microscope stage |
| | smear the cell sample onto a microscope slide |

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

**(1)**

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

**(1)**

.....

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(iv) Give two ways that the student could avoid health hazards when working with living cells.

**(2)**

1 .....

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2 .....

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# 4. Question 3 · part b.i · 1 mark

November 2020 Paper 1R

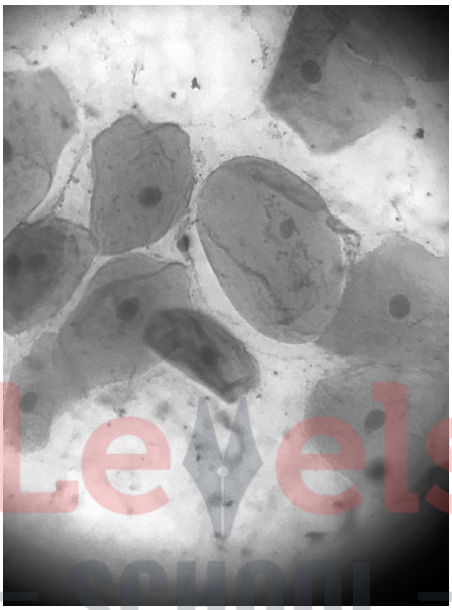
Further practice · Attempt part b.i only.

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(b) The photograph shows an image of a cheek cell seen under a light microscope using a 10× eyepiece lens and a 40× objective lens.



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© Esmiles/Shutterstock

(i) Calculate the total magnification used to view the cheek cell.

**(1)**

total magnification = .....

(ii) The actual diameter of the cheek cell shown in the photograph is 0.018 mm.

Convert the diameter of the cheek cell into micrometre (μm).
[1 μm = $10^{-6}$ m]

**(2)**

diameter = ..... μm

**(Total for Question 3 = 9 marks)**

## 4. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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# 5. Question 5 · part c · 4 marks

June 2021 Paper 1

Further practice · Attempt part c only.

**5** The diagram shows a sperm.




(a) (i) Describe the function of a sperm.

**(2)**

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

**(2)**

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

**(3)**

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## 5. Question 5 · part c · 4 marks

June 2021 Paper 1

Further practice · Attempt part c only.

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(c) The actual length of the sperm between X and Y on the diagram is 3 µm.

Calculate the magnification of the sperm shown in the diagram.

[1000 µm = 1 mm]

**(4)**

magnification = × .....

**(Total for Question 5 = 11 marks)**

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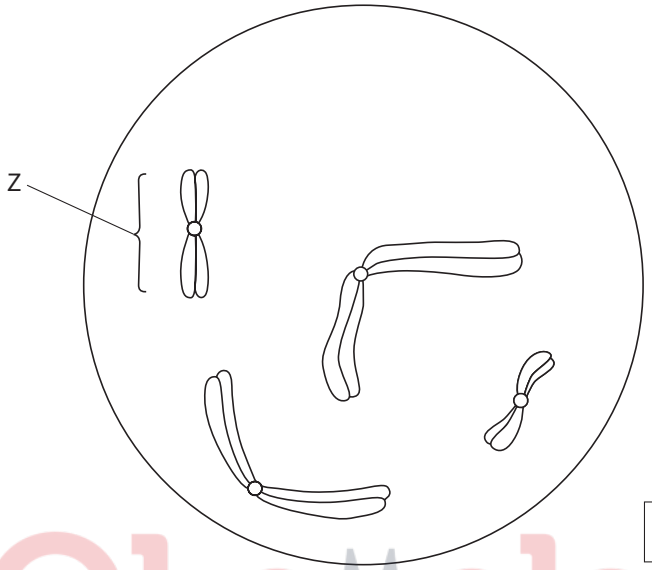
Working and answer

## 6. Question 7 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

**7** The diagram shows a cell in early stages of mitosis.




(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

**(2)**

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

**(3)**

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## 6. Question 7 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = $1 \times 10^{-6}$ m]

(3)

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length of structure Z = ..... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

.....

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**(Total for Question 7 = 11 marks)**

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## 6. Question 7 · part b.ii · 3 marks · Working and answer

Working and answer

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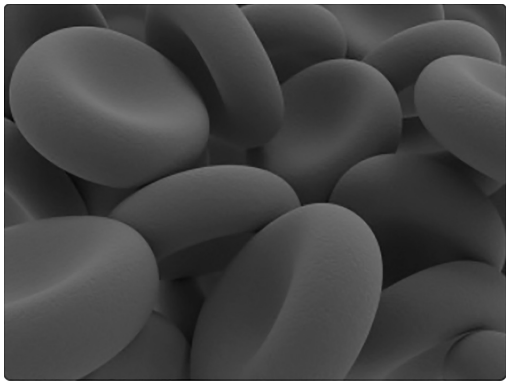
# 7. Question 1 · part b.iii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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**1** (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

**(3)**

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(ii) Explain how the structure of the red blood cell is adapted to its function.

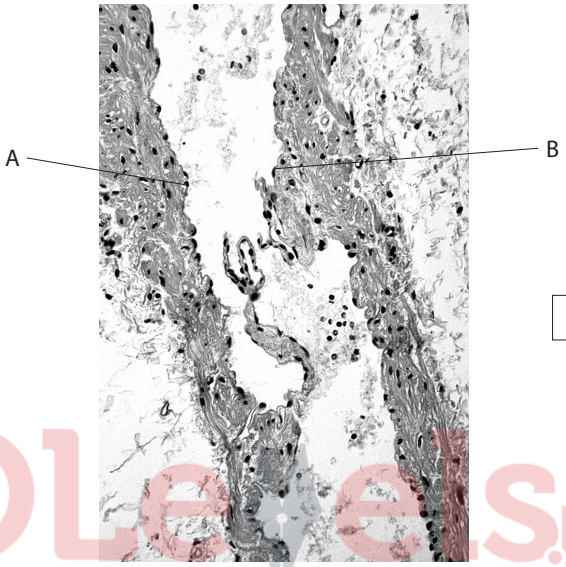
**(4)**

# 7. Question 1 · part b.iii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

(b) The photograph shows a section through a blood vessel in a leg.




magnification ×10

(Source: © DR. GLADDEN WILLIS, VISUALS UNLIMITED/SCIENCE PHOTO LIBRARY)

(i) State the type of blood vessel shown in the photograph.

**(1)**

(ii) Explain your choice of blood vessel.

**(2)**

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## 7. Question 1 · part b.iii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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(iii) Calculate the actual distance between points A and B.

**(2)**

distance = ..... mm

**(Total for Question 1 = 12 marks)**

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Working and answer

# 8. Question 1 · part a.ii · 3 marks

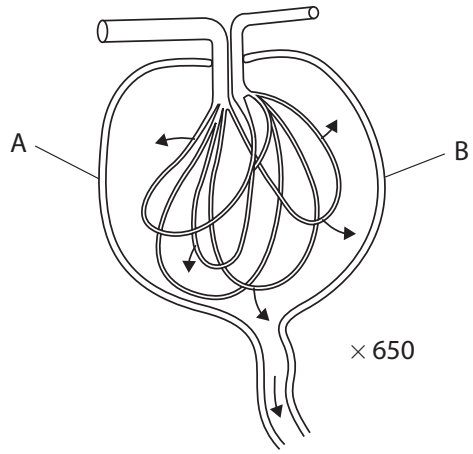
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Further practice · Attempt part a.ii only.

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**1** (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).




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(i) Complete the diagram to show the missing parts of the kidney tubule.

**(3)**

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

**(3)**

width of Bowman's capsule = ..... mm

# 8. Question 1 · part a.ii · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

**(2)**

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

**(1)**

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 $dm^3$ of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

| Time after drinking in hours | Volume of urine in $cm^3$ | |
|---|---|---|
| | day 1 (20 °C) | day 2 (30 °C) |
| 0.5 | 70 | 50 |
| 1.5 | 175 | 70 |
| 2.0 | 300 | 200 |
| 2.5 | 250 | 190 |
| 3.0 | 210 | 150 |
| 4.0 | 100 | 50 |

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## 8. Question 1 · part a.ii · 3 marks · Working and answer

Working and answer

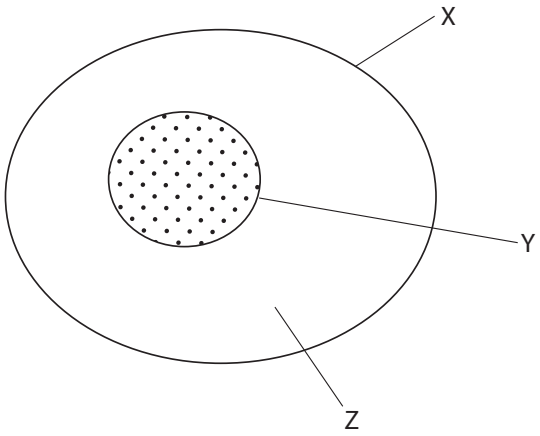
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# 9. Question 1 · part c.ii · 3 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

**1** (a) The diagram shows a human body cell.




(i) Name the parts X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

**(1)**

.....

(iii) Explain how body cells are organised to form organs.

**(2)**

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## 9. Question 1 · part c.ii · 3 marks

June 2022 Paper 1R

Further practice · Attempt part c.ii only.

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(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

**(3)**

| Process | Definition |
|---|---|
| | requires energy to move molecules against a concentration gradient |
| osmosis | |
| | water moves from a high water potential to a low water potential |
| diffusion | |
| | water moves from a low water potential to a high water potential |
| active transport | |
| | molecules move down a concentration gradient |

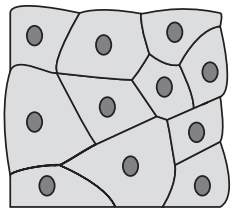
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# 9. Question 1 · part c.ii · 3 marks

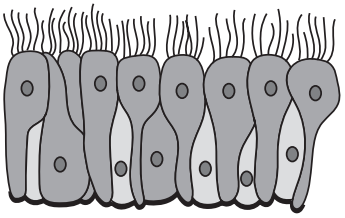
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Further practice · Attempt part c.ii only.

(c) (i) The diagram shows two types of epithelial tissue, A and B.



Tissue A



Tissue B

(Source: © Timonina/Shutterstock)

Give the name of each type of epithelial tissue.

**(2)**

A .....

B .....

(ii) An epithelial cell has an actual width of 60 micrometres (μm).

The cell is magnified 40 times.

Calculate the image width of this cell in mm.

[1 mm = 1000 μm]

**(3)**

image width = ..... mm

**(Total for Question 1 = 14 marks)**

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## 9. Question 1 · part c.ii · 3 marks · Working and answer

Working and answer

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## 10. Question 4 · part a.ii · 2 marks

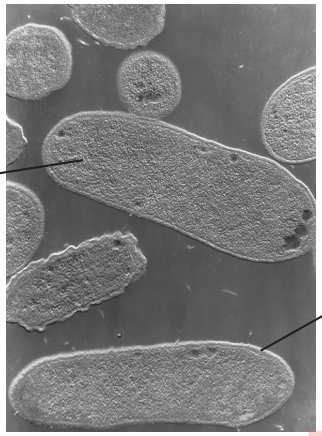
January 2023 Paper 1

Further practice · Attempt part a.ii only.

**4** Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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# 10. Question 4 · part a.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

**(1)**

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

**(3)**

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## 10. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

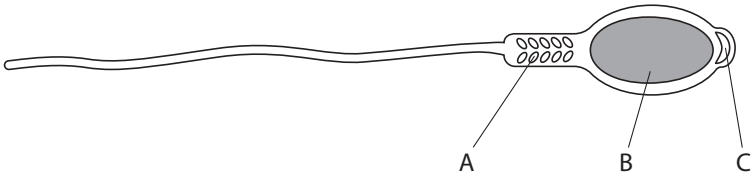
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# 11. Question 6 · part a.iii · 4 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

**6** The diagram shows a human sperm.




(a) (i) Name the parts A, B and C.

**(3)**

A .....

B .....

C .....

(ii) Describe the function of part C.

**(3)**

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(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

**(4)**

magnification = .....

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## 11. Question 6 · part a.iii · 4 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

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(b) Describe the function of the structure labelled A in the process of fertilisation.

**(3)**

**(Total for Question 6 = 13 marks)**

Working and answer

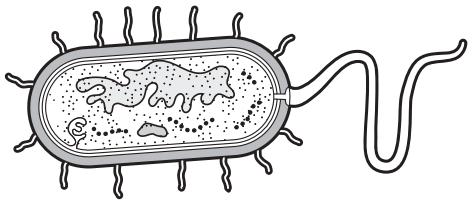
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# 12. Question 1 · part b.ii · 3 marks

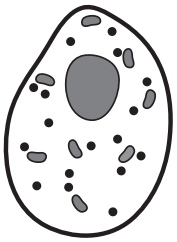
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Further practice · Attempt part b.ii only.

**1** (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

**(2)**

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

**(2)**

1 .....

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2 .....

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# 12. Question 1 · part b.ii · 3 marks

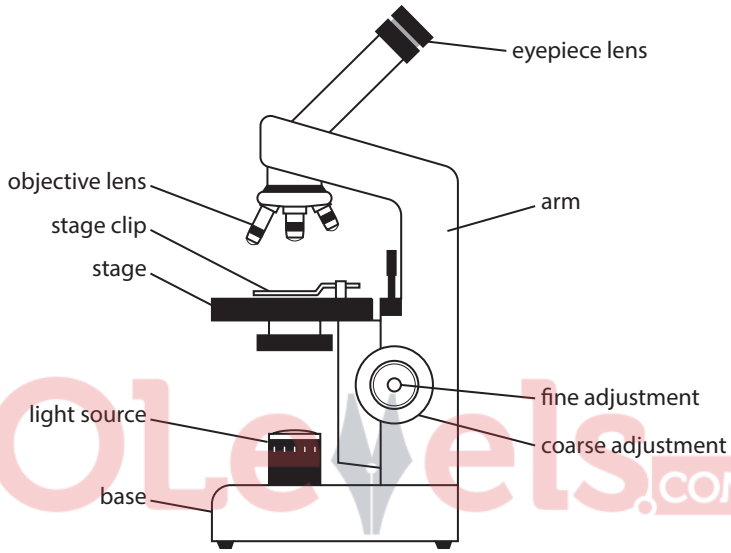
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Further practice · Attempt part b.ii only.

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(b) Many structures in cells can be viewed more clearly using a light microscope.

The diagram shows a light microscope.




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(i) Draw **one** straight line from each part of the light microscope to its function.

**(3)**

| Part of microscope | Function |
|---|---|
| objective lens | holds the microscope slide in place |
| stage clip | moves the stage to produce a clear, focused image |
| fine adjustment | magnifies the sample |
| | keeps the microscope steady |

# 12. Question 1 · part b.ii · 3 marks

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

(ii) The image size of a bacterial cell viewed using an electron microscope is 5.0 cm.

The bacterial cell has been magnified 50 000 times.

Calculate the actual size of this bacterial cell in mm.

image size = actual size × magnification

**(3)**

actual size of bacterial cell = ..... mm

**(Total for Question 1 = 10 marks)**

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Working and answer

## 13. Question 4 · part b.ii · 3 marks

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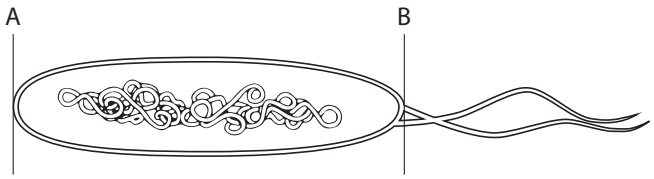
Further practice · Attempt part b.ii only.

**4** (a) State what is meant by the term **pathogen**.

**(1)**

.....

(b) The diagram shows a pathogenic bacterium.




(i) Label **three** structures shown on the diagram.

**(3)**

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

**(3)**

magnification = × .....

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## 13. Question 4 · part b.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part b.ii only.

(c) The effectiveness of a vaccine in preventing disease was studied over six months.

Two different groups of people were used. One group of people were aged between 45 and 64 years and the second group were aged over 65 years.

The results are shown in the table.

| | Percentage effectiveness against disease (%) | |
|---|---|---|
| **Time since vaccination in months** | **45 to 64 years** | **over 65 years** |
| 1 | 90 | 80 |
| 2 | 85 | 78 |
| 3 | 80 | 75 |
| 4 | 68 | 58 |
| 5 | 60 | 50 |
| 6 | 50 | 40 |

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## 13. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

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